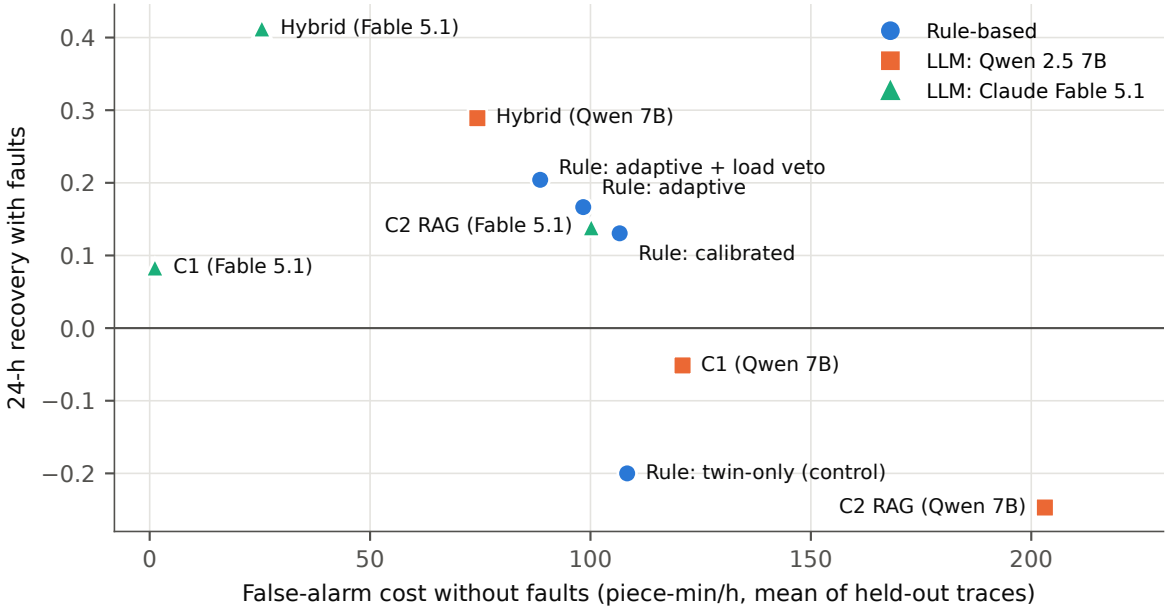
Rule-based
LLM: Qwen 2.5 7B
LLM: Claude Fable 5.1
Hybrid (Fable 5.1)
Hybrid (Qwen 7B)
Rule: adaptive + load veto
Rule: adaptive
C2 RAG (Fable 5.1)
Rule: calibrated
C1 (Fable 5.1)
C1 (Qwen 7B)
Rule: twin-only (control)
C2 RAG (Qwen 7B)
24-h recovery with faults
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
0
50
100
150
200
False-alarm cost without faults (piece-min/h, mean of held-out traces)